

Divya Rajparia

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Objective

To obtain a job opportunity that will allow me to gain real-world experience while leveraging the strong academic skills I have acquired through my undergraduate program.

Education Details

Indian Institute of Technology, Hyderabad

Oct 2022 - Present

BTech in Engineering Science with Specialization in Computer Science

- GPA: 9.33/10
- Ranked 2nd in my department

PACE Junior Science College, Mumbai

May 2020 - Aug 2022

High School Diploma in Math, Science, and Computer Science

- JEE Mains 2022: scored **99.90 percentile** among over **1 million** students
- Secured **91.50%** in High School (12th Grade)

Lokhandwala Foundation School, Mumbai

June 2009 - May 2020

High School Diploma in Math, Science, and Computer Science

- Completed 10th standard with **98.17%**
- Silver Medalist, Dr. Homi Bhabha Young Scientist Competition (In the top **0.45%** students).
- Senior Diploma Holder in Music (Keyboard)

Experience

Enhancing tumor segmentation using Synthetic Data

Los Angeles, USA

University of Southern California

June 2025 - Present

Currently working as a research intern at the Laboratory for Machine Learning, Health, and Biomedicine, where we are developing methods to automate the complex task of **tumor segmentation** in breast and lung cancer from ultrasound, CT, and MRI scans. Manual annotation of images is time-consuming for clinicians, and automating this process can significantly enhance diagnostic efficiency.

- One of the major challenges is the domain shift that occurs when models trained on one dataset underperform on others due to differences in imaging conditions. To address this, we are leveraging **cross-domain style transfer** techniques to harmonize features across datasets, enabling better generalization.
- Additionally, we are exploring **federated learning** to collaboratively train segmentation models across institutions without sharing sensitive patient data, ensuring both privacy and robustness. Our goal is to build a domain-adaptive, federated, and semantically rich segmentation model that generalizes well across diverse clinical scenarios.

Making Vision Transformers More Human Like: Learning from Parts

Hyderabad, India

Lab for Vision and Image Analysis, IIT Hyderabad (Submitted to Neurips 2025)

June 2024 - May 2025

Vision Transformers (ViTs) are a cutting-edge approach to image recognition that adapt transformer models (from language processing) to visual data. We investigated how Vision Transformers (ViTs) **understand complex images** by studying how they combine visual parts to form complete representations. This was driven by the need to make **ViTs more interpretable** and explainable, especially as they're increasingly used across computer vision tasks.

- Designed a framework using wavelet transforms to break images into meaningful pieces (frequency sub-bands) and tested whether ViTs can linearly reconstruct the original from these parts.
- Built and trained models in PyTorch, ran large-scale experiments on ImageNet, and analyzed how well ViTs retain compositional information.
- We demonstrated that ViTs, which are highly non-linear, do **exhibit compositional behavior** at deeper layers. Simple linear combinations of wavelet primitives can closely approximate the original

representations.

- We also tested our model under **real-world distortions**, such as additive noise and image compression, and found out that our framework persists even under such conditions.

DES: A Smarter Way to Grow Machine Learning Models Over Time

Hyderabad, India

Machine Learning and Vision Lab, IIT Hyderabad

Jan 2025 – May 2025

Continual Learning is a Machine Learning paradigm where we train models on tasks that arrive over time - like how humans learn - without forgetting past knowledge. Most existing methods rely on unrealistic assumptions (balanced data, access to old samples), which limit their use in real-world settings with privacy concerns or skewed data. We addressed this gap with a more practical approach:

- Proposed a new learning setup called AFCIL (**A**ssumption-**F**ree **C**lass-**I**ncremental **L**earning) that removes these assumptions - no need for stored data, works with imbalanced data, and supports pre-trained models like CLIP.
- Designed and implemented a novel method, **Dynamic Expert Saturation** (DES), which splits learning into two simple stages. The first stage focuses on learning the common (head) classes, and the second stage incrementally (dynamically) saturates the model to the rare (tail) classes. This avoids catastrophic forgetting and handles class imbalance naturally.
- Used **PyTorch** and **CLIP** backbone with lightweight adapters (MLPs), allowing efficient training without changing the base model. Final predictions were made by combining (weighting) the expert outputs.
- Achieved state-of-the-art results on CIFAR100-LT and ImageNet-LT, outperforming all existing methods including DualPrompt, CODA, and MoE-Adapter by up to 6% in accuracy.

Skills and Courses

- Fundamental Sciences: Life Sciences, Cognitive Science, Psychopathology, Biological Engineering
- Programming in Python, C/C++, MATLAB, HTML/CSS, JavaScript
- ML frameworks: Scikit-Learn, TensorFlow, and PyTorch
- Advanced Mathematics: Calculus, Linear Algebra, Statistics, Probability
- Computer Science: Computer Networks, Computer Architecture, Operating Systems, Compilers

Societies and Associations

Ideation Head, Entrepreneurship Cell IIT Hyderabad

May 2023 - Sept 2024

As the Head of Ideation at E-Cell, I was involved in hosting various entrepreneurial events. We brainstormed the execution plan of the E-Summit, convinced eminent speakers to attend our events, and hosted networking sessions for aspiring entrepreneurs.

Head, Vibes – the music club of IIT Hyderabad

May 2023 - May 2024

As the head of the Music Club, I had the responsibility of managing the club and promoting a culture of Music on our campus. I oversaw all aspects of the club like **finances, events, and media relations**. I also **interviewed candidates** for induction to the club. This experience enhanced my **communication skills** and **ability to work with teams**.

Team Member, IIT Hyderabad Athletics Team

Personal Details

- Country of Citizenship – United States of America (USA)